Foundational and Canonical References

Theory and practice of obtaining, processing and evaluating satellite data for agriculture and soil health | October 2026

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Foundational theory and textbooks (open access unless noted)

[1] Cardille, J.A., Crowley, M.A., Saah, D. & Clinton, N.E. (eds.) (2024). *Cloud-Based Remote Sensing with Google Earth Engine: Fundamentals and Applications*. Springer, 1,247 pp. Open access (CC BY). https://doi.org/10.1007/978-3-031-26588-4 - 55 lab-style chapters from image basics to regression, time-series, classification and agricultural applications.

[2] Canada Centre for Mapping and Earth Observation (Natural Resources Canada). *Fundamentals of Remote Sensing* tutorial. Free online primer on electromagnetic radiation, sensors, platforms and image analysis. https://natural-resources.canada.ca/maps-tools-and-publications/satellite-imagery-and-air-photos/tutorial-fundamentals-remote-sensing

[3] NASA Applied Remote Sensing Training (ARSET). *Fundamentals of Remote Sensing* and *Agricultural Crop Classification with Synthetic Aperture Radar and Optical Remote Sensing*. Free recorded courses and exercises. https://appliedsciences.nasa.gov/get-involved/training

[4] Jensen, J.R. (2015). *Introductory Digital Image Processing: A Remote Sensing Perspective*, 4th ed. Pearson. (Textbook; not open access.) Standard reference for radiometric/geometric correction and classification.

[5] Lillesand, T., Kiefer, R.W. & Chipman, J. (2015). *Remote Sensing and Image Interpretation*, 7th ed. Wiley. (Textbook; not open access.)

[6] McBratney, A.B., Mendonca Santos, M.L. & Minasny, B. (2003). On digital soil mapping. *Geoderma* 117, 3-52. https://doi.org/10.1016/S0016-7061(03)00223-4 - defines the SCORPAN framework used by all satellite-covariate soil mapping.

[7] Mulder, V.L., de Bruin, S., Schaepman, M.E. & Mayr, T.R. (2011). The use of remote sensing in soil and terrain mapping - a review. *Geoderma* 162, 1-19. https://doi.org/10.1016/j.geoderma.2010.12.018

Canonical reviews of satellite remote sensing in agriculture

[1] Weiss, M., Jacob, F. & Duveiller, G. (2020). Remote sensing for agricultural applications: a meta-review. *Remote Sensing of Environment* 236, 111402. https://doi.org/10.1016/j.rse.2019.111402 - the central synthesis of techniques, retrievable variables and applications (open-access author version available via HAL/IRD).

[2] Sishodia, R.P., Ray, R.L. & Singh, S.K. (2020). Applications of remote sensing in precision agriculture: a review. *Remote Sensing* 12, 3136. https://doi.org/10.3390/rs12193136 (open access).

[3] Khanal, S., KC, K., Fulton, J.P., Shearer, S. & Ozkan, E. (2020). Remote sensing in agriculture - accomplishments, limitations, and opportunities. *Remote Sensing* 12, 3783. https://doi.org/10.3390/rs12223783 (open access).

[4] Segarra, J., Buchaillot, M.L., Araus, J.L. & Kefauver, S.C. (2020). Remote sensing for precision agriculture: Sentinel-2 improved features and applications. *Agronomy* 10, 641. https://doi.org/10.3390/agronomy10050641 (open access).

[5] Wu, B. et al. (2023). Challenges and opportunities in remote sensing-based crop monitoring: a review. *National Science Review* 10, nwac290. https://doi.org/10.1093/nsr/nwac290 (open access).

[6] Sentinel-2 for crop yield estimation: a systematic review (2026). *Smart Agricultural Technology*; preprint arXiv:2603.23779 (open access).

Sensors, indices and physical retrieval

[1] Tucker, C.J. (1979). Red and photographic infrared linear combinations for monitoring vegetation. *Remote Sensing of Environment* 8, 127-150. https://doi.org/10.1016/0034-4257(79)90013-0 - origin of NDVI.

[2] Huete, A. et al. (2002). Overview of the radiometric and biophysical performance of the MODIS vegetation indices. *Remote Sensing of Environment* 83, 195-213. https://doi.org/10.1016/S0034-4257(02)00096-2

[3] Drusch, M. et al. (2012). Sentinel-2: ESA's optical high-resolution mission for GMES operational services. *Remote Sensing of Environment* 120, 25-36. https://doi.org/10.1016/j.rse.2011.11.026

[4] Torres, R. et al. (2012). GMES Sentinel-1 mission. *Remote Sensing of Environment* 120, 9-24. https://doi.org/10.1016/j.rse.2011.05.028

[5] Roy, D.P. et al. (2014). Landsat-8: science and product vision for terrestrial global change research. *Remote Sensing of Environment* 145, 154-172. https://doi.org/10.1016/j.rse.2014.02.001 (open access).

[6] Claverie, M. et al. (2018). The Harmonized Landsat and Sentinel-2 surface reflectance data set. *Remote Sensing of Environment* 219, 145-161. https://doi.org/10.1016/j.rse.2018.09.002

[7] Entekhabi, D. et al. (2010). The Soil Moisture Active Passive (SMAP) mission. *Proceedings of the IEEE* 98, 704-716. https://doi.org/10.1109/JPROC.2010.2043918

[8] Jacquemoud, S. et al. (2009). PROSPECT + SAIL models: a review of use for vegetation characterization. *Remote Sensing of Environment* 113, S56-S66. https://doi.org/10.1016/j.rse.2008.01.026

[9] Main-Knorn, M. et al. (2017). Sen2Cor for Sentinel-2. *Proc. SPIE* 10427, Image and Signal Processing for Remote Sensing XXIII. https://doi.org/10.1117/12.2278218

[10] Zhu, Z. & Woodcock, C.E. (2012). Object-based cloud and cloud shadow detection in Landsat imagery (Fmask). *Remote Sensing of Environment* 118, 83-94. https://doi.org/10.1016/j.rse.2011.10.028

Soil properties from satellite data (SOC, pH, CEC, N, texture)

[1] Angelopoulou, T., Tziolas, N., Balafoutis, A., Zalidis, G. & Bochtis, D. (2019). Remote sensing techniques for soil organic carbon estimation: a review. *Remote Sensing* 11, 676. https://doi.org/10.3390/rs11060676 (open access) - includes an introduction to soil spectroscopy.

[2] Lima, A.A.J. et al. (2025). Soil organic carbon assessment using remote-sensing data and machine learning: a systematic literature review. *Remote Sensing* 17, 882. https://doi.org/10.3390/rs17050882 (open access).

[3] Vaudour, E. et al. (2019). Sentinel-2 image capacities to predict common topsoil properties of temperate and Mediterranean agroecosystems. *Remote Sensing of Environment* 223, 21-33. https://doi.org/10.1016/j.rse.2019.01.006 - PLSR for SOC, pH, CEC, texture, CaCO3, iron.

[4] Castaldi, F. et al. (2019). Evaluating the capability of the Sentinel 2 data for soil organic carbon prediction in croplands. *ISPRS Journal of Photogrammetry and Remote Sensing* 147, 267-282. https://doi.org/10.1016/j.isprsjprs.2018.11.026

[5] Rogge, D. et al. (2018). Building an exposed soil composite processor (SCMaP) for mapping spatial and temporal characteristics of soils with Landsat imagery (1984-2014). *Remote Sensing of Environment* 205, 1-17. https://doi.org/10.1016/j.rse.2017.11.004

[6] Dematte, J.A.M. et al. (2018). Geospatial Soil Sensing System (GEOS3): a powerful data mining procedure to retrieve soil spectral reflectance from satellite images. *Remote Sensing of Environment* 212, 161-175. https://doi.org/10.1016/j.rse.2018.04.047

[7] Zizala, D., Minarik, R. & Zadorova, T. (2019). Soil organic carbon mapping using multispectral remote sensing data: prediction ability of data with different spatial and spectral resolutions. *Remote Sensing* 11, 2947. https://doi.org/10.3390/rs11242947 (open access).

[8] Wang, S. et al. (2022). Using soil library hyperspectral reflectance and machine learning to predict soil organic carbon: assessing potential of airborne and spaceborne optical soil sensing. *Remote Sensing of Environment* 271, 112914. https://doi.org/10.1016/j.rse.2022.112914

[9] Poggio, L., de Sousa, L.M., Batjes, N.H., Heuvelink, G.B.M., Kempen, B., Ribeiro, E. & Rossiter, D. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *SOIL* 7, 217-240. https://doi.org/10.5194/soil-7-217-2021 (open access).

[10] Orgiazzi, A., Ballabio, C., Panagos, P., Jones, A. & Fernandez-Ugalde, O. (2018). LUCAS Soil, the largest expandable soil dataset for Europe: a review. *European Journal of Soil Science* 69, 140-153. https://doi.org/10.1111/ejss.12499 (open access).

[11] Satellite remote sensing enables monitoring of soil organic carbon decline in croplands of Jilin, China (2026). Open access via PubMed Central (PMC12917285) - compares PLSR SOC models across MODIS, Landsat OLI and Sentinel-2 MSI and proposes the Ratio Soil Index.

Soil biology and microbial inference from remote sensing (emerging)

[1] eDNA biodiversity from space: predicting soil bacteria and fungi alpha diversity in forests using DESIS satellite remote sensing (2025). *International Journal of Remote Sensing*. https://doi.org/10.1080/01431161.2025.2464958 (open access).

[2] Exploring crop health and its associations with fungal soil microbiome composition using machine learning applied to remote sensing data (2025). *Communications Earth & Environment*. https://doi.org/10.1038/s43247-025-02330-0 (open access) - NDVI adjusted with LUCAS topsoil and ERA5 covariates, linked to fungal taxa.

[3] Bastida, F. et al. (2021). Soil microbial diversity-biomass relationships are driven by soil carbon content across global biomes. *The ISME Journal* 15, 2081-2091. https://doi.org/10.1038/s41396-021-00906-0 (open access) - explains why SOC and pH maps carry information about microbial communities.

[4] Joergensen, R.G. (2024). A hitchhiker's guide: estimates of microbial biomass and microbial gene abundance in soil. *Biology and Fertility of Soils*. https://doi.org/10.1007/s00374-024-01810-3 (open access) - ground-truth methods for biomass and bacteria:fungi ratios.

[5] Frostegard, A., Tunlid, A. & Baath, E. (2011). Use and misuse of PLFA measurements in soils. *Soil Biology and Biochemistry* 43, 1621-1625. https://doi.org/10.1016/j.soilbio.2010.11.021 - the reference method for fungal:bacterial ratios used to calibrate any remote-sensing proxy.

Platforms, machine learning and evaluation methodology

[1] Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D. & Moore, R. (2017). Google Earth Engine: planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment* 202, 18-27. https://doi.org/10.1016/j.rse.2017.06.031 (open access).

[2] Belgiu, M. & Dragut, L. (2016). Random forest in remote sensing: a review of applications and future directions. *ISPRS Journal of Photogrammetry and Remote Sensing* 114, 24-31. https://doi.org/10.1016/j.isprsjprs.2016.01.011

[3] Meyer, H., Reudenbach, C., Wollauer, S. & Nauss, T. (2019). Importance of spatial predictor variable selection in machine learning applications - moving from data reproduction to spatial prediction. *Ecological Modelling* 411, 108815. https://doi.org/10.1016/j.ecolmodel.2019.108815 - spatial cross-validation.

[4] Meyer, H. & Pebesma, E. (2021). Predicting into unknown space? Estimating the area of applicability of spatial prediction models. *Methods in Ecology and Evolution* 12, 1620-1633. https://doi.org/10.1111/2041-210X.13650 (open access).

[5] Olofsson, P. et al. (2014). Good practices for estimating area and assessing accuracy of land change. *Remote Sensing of Environment* 148, 42-57. https://doi.org/10.1016/j.rse.2014.02.015 - standard for accuracy assessment of crop/land-cover maps.

[6] Bellon-Maurel, V. et al. (2010). Critical review of chemometric indicators commonly used for assessing the quality of the prediction of soil attributes by NIR spectroscopy. *TrAC Trends in Analytical Chemistry* 29, 1073-1081.

https://doi.org/10.1016/j.trac.2010.05.006 - RPD/RPIQ interpretation.